

ACTIVITY 7 — FACILITATOR GUIDE

Pareto, Run Charts, Variation, Yield, DPU, and DPMO

TYPE	DURATION	DMAIC PHASE	MAPS TO
Core Lab	75 min	ANALYZE	A3, A4

Purpose of this activity

Analyse process performance data to quantify and prioritise (A3, A4).

Turn collected data into decisions. Apply the Pareto principle to find the vital few categories, use a run chart to see behaviour over time, and calculate the standard Six Sigma process metrics.

What learners produce

A Pareto chart, a run chart with the pattern named, and calculated yield, DPU, DPO, DPMO and sigma level that reconcile with the data.

Materials required

- Learner Worksheet (one per learner)
- Checklist (one per group)
- A laptop with Excel, Google Sheets or LibreOffice Calc
- The CSV files in this activity's data/ and templates/ folders
- Flipchart or A3 sheet and markers
- The course slides for the scenario and the framework
- A browser — this activity uses an online tool (URL in the run-sheet).

Set-up

Groups of 3-5. Distribute one worksheet per learner and one checklist per group. Every activity runs on the same continuous Contoso Service Desk scenario, so remind learners where this one sits in that story before starting.

Scenario / briefing material

Contoso Service Desk handles employee IT requests. Employees complain that tickets take too long to be assigned, that they must keep chasing for status, and that agents apply different definitions of what counts as "resolved". Handoffs between the front line and the specialist queues are unclear, and although ticket data exists it is inconsistent. You are supporting the improvement team as a Yellow Belt — you do not lead the project.

Two weeks of data have been collected: 400 tickets, of which 96 contained at least one defect, with 120 defects recorded in total across 6 defect opportunities per ticket. Mean assignment time is 55 minutes (median 46) against a 30-minute improvement goal. The CSV files in this activity's data/ folder are that extract.

Files in this activity folder

data/ — the mock data you analyse:

- assignment-times.csv — 400 rows (date, ticket_id, assignment_time_min)
- daily-summary.csv — 10 rows (date, tickets, mean_assignment_min, median_assignment_min, defects, missed_30min_goal)
- defect-counts.csv — 6 rows (defect_category, count)
- service-desk-tickets.csv — 400 rows (ticket_id, date, ticket_type, channel, assigned_queue, agent, assignment_time_min, met_30min_goal, defect_count, defect_categories, reopened)

templates/ — the worksheets you complete:

- pareto-table.csv (defect_category, count, percent, cumulative_percent)
- process-metrics.csv (metric, formula, your_calculation)

Open these in Excel, Google Sheets or LibreOffice Calc. Work on a copy so the originals stay clean for revision.

Upload data/assignment-times.csv to NovaSPC for the run chart. Worked answers are in solution/pareto-table-answers.csv and solution/process-metrics-answers.csv — attempt the lab before opening them.

How to run it — step by step

#	Instruction to learners	Facilitator note
1	Build the Pareto table: category, count, percentage and cumulative percentage.	The cumulative column must reach exactly 100% — a rounding drift means a lost defect.
2	In your group, open the collaborative Pareto tool — one member creates the session and shares the code, then everyone joins, brainstorms and votes to produce the live Pareto chart. Tool: https://alfredang.github.io/paretochart/	The run chart hides a genuine SHIFT from day 7 — do not let learners call it a trend.
3	Identify the vital few categories driving about 80% of the problem.	Insist DPMO is stated WITH the sigma level it produced; a bare number proves nothing.
4	Export your assignment-time data as CSV and plot a run chart in NovaSPC. Tool: https://alfredang.github.io/novaspc/	Answers are in solution/ — do not release them before the group has attempted it.
5	Interpret the run chart for trend, shift, cluster and oscillation patterns.	
6	Calculate yield, DPU, DPO and DPMO from your defect data.	
7	Convert DPMO to a sigma level and interpret what it says about the process.	
8	Distinguish common-cause from special-cause variation and why the response differs.	

Suggested timing

Minutes	Phase
15 min	Build the Pareto table from defect-counts.csv
15 min	Run the collaborative Pareto session
10 min	Identify the vital few
15 min	Plot the run chart in NovaSPC
10 min	Interpret the run chart patterns
10 min	Calculate yield, DPU, DPO, DPMO and sigma

Total: 75 minutes

Discussion & decision prompts

1. Which two categories make up the vital few, and what percentage do they carry?
2. What pattern does the run chart show — trend, shift, cluster or oscillation?
3. What can you NOT yet conclude from the Pareto alone?

Debrief

Bring the class back together and draw out the learning:

- Ask two or three groups to show what they produced, not just what they concluded.
- Ask what surprised them — the gap between what they assumed and what the data showed.
- Link the activity explicitly back to the ANALYZE phase of DMAIC.
- Remind learners this activity rehearses what the Case Study assessment will require.

Reflection questions:

- What would a sigma level of ~3.1 mean if it were your own process?
- Where in your work do you react to a single data point instead of the pattern?

Facilitator notes — what to watch for

- The cumulative column must reach exactly 100% — a rounding drift means a lost defect.
- The run chart hides a genuine SHIFT from day 7 — do not let learners call it a trend.
- Insist DPMO is stated WITH the sigma level it produced; a bare number proves nothing.
- Answers are in solution/ — do not release them before the group has attempted it.

Success criterion

Your cumulative percentage column reaches 100%, and you can state the sigma level with the DPMO it came from.

Assessment link

This activity builds the ANALYZE phase of DMAIC. All activities are in-class activities assessed indirectly through the open-book Case Study (CS), which reuses the same techniques on the same Contoso Service Desk scenario and the same figures (400 tickets, yield 76%, DPMO 50,000). Knowledge is assessed in the Written Assessment (WA-SAQ). Nothing is assessed that is not practised here or taught in the slides.